

Scenario 1: Flight Delay Analysis

Question:

An airline tracks flight delays (in minutes) for 20 flights. How do you analyze the flight delays to calculate percentiles, detect outliers, and evaluate the overall distribution?

Answer:

- **Percentiles Calculation:**
Calculate the 10th, 25th, 50th (Median), 75th, and 90th percentiles to understand how flight delays are distributed at different levels.
- **IQR Calculation:**
Compute **IQR = Q3 - Q1**. Detect outliers using the formula:
 - Outliers are values outside $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$.
- **Distribution:**
Use a **box plot** to visualize the spread, median, quartiles, and potential outliers. A **histogram** helps assess the distribution shape, skewness, and frequency of delays.

Scenario 2: Employee Salary Analysis

Question:

A company wants to analyze the salary distribution of its employees to understand the central tendency and determine if the data is skewed. How should this be done?

Answer:

- **Central Tendency Calculation:**
Calculate **Mean**, **Median**, and **Mode** to understand the central value of the salaries.
- **Skewness Check:**
 - If **Mean > Median**, the data is **Right Skewed** (higher salaries skew the mean).
 - If **Mean < Median**, the data is **Left Skewed** (lower salaries pull the mean down).
 - If **Mean $\approx$ Median**, the data is **Symmetrical**.
- **Best Representation:**
Use the **Median** when there are outliers, as it better represents the central tendency in the presence of extreme values.

Scenario 3: Product Sales Analysis

Question:

A retail store records product sales over 15 days. How do you create a frequency distribution table and visualize the sales data?

Answer:

- **Frequency Distribution:**
Divide the sales data into intervals (e.g., 5 or 10 units). Count the sales within each interval to understand how sales are distributed.
- **Visualization:**
 - **Histogram:** Displays the frequency distribution across sales intervals.
 - **Bar Plot:** Shows trends in sales, helping to visualize changes over time.

Scenario 4: Student Exam Performance Analysis

Question:

A school wants to analyze the exam performance of students across three subjects: Mathematics, Science, and English. How can Data Science concepts be applied to understand their performance?

Answer:

1. Data Preprocessing:

- Handle missing values by imputing with the mean or median.
- Convert categorical values (if any) using label encoding or one-hot encoding.

2. Descriptive Statistics:

- Calculate mean, median, mode, and standard deviation for each subject to understand the central tendency and dispersion.

3. Visualization:

- Use box plots to detect outliers.
- Create histograms to visualize the distribution of scores.
- Plot scatter plots to check relationships between subjects.

4. Correlation Analysis:

- Compute correlation coefficients to see how subject scores relate to each other.
- Use a heatmap to visualize correlations.

Scenario 5: Clinical Trial for Diabetes Medication

Question:

A pharmaceutical company conducted a clinical trial with two groups: one receiving medication and the other a placebo. How do you perform a hypothesis test to determine the effectiveness of the medication?

Answer:

- **Hypothesis Test:**
 - **H_0 (Null Hypothesis):** No difference between medication and placebo.
 - **H_1 (Alternative Hypothesis):** Medication lowers blood sugar more than the placebo.
- **T-Test:**
 - If $p \leq 0.05$, reject **H_0** (indicating medication is effective).
 - If $p > 0.05$, fail to reject **H_0** (no significant difference).